

Package

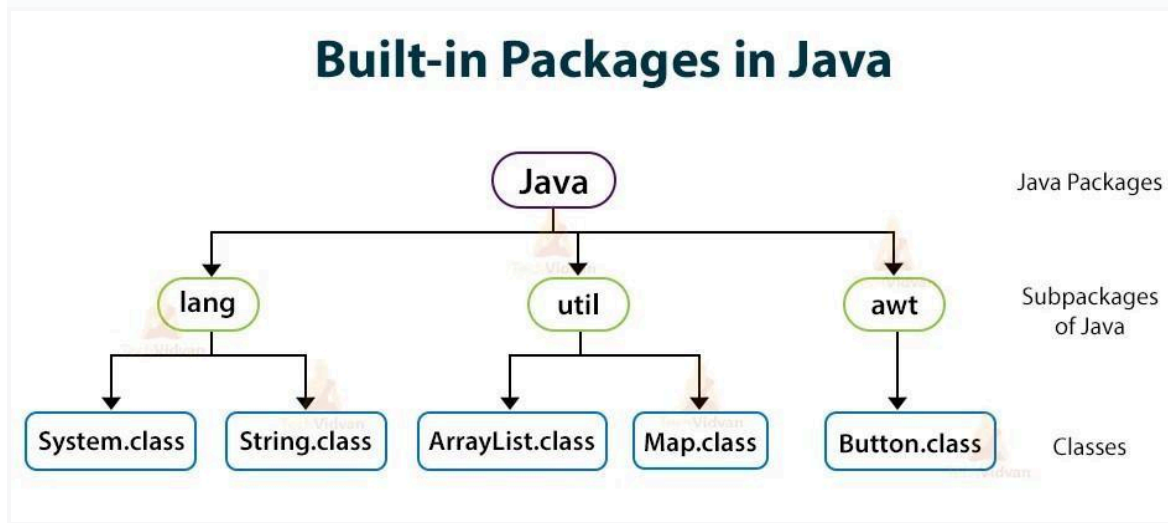
A Java package is a collection of similar types of sub-packages, interfaces, and classes. They help you manage and group related classes, interfaces, and sub-packages to avoid naming conflicts and create a more organized and maintainable codebase.

Example:

Directories or folders on your computer's file system(manage files).

In Java, there are two types of packages: built-in packages and user-defined packages.

in-built packages : They are available in Java, including util, lang, awt etc. We can import all members of a package using package name.* statement



java.lang is a special package that is automatically **imported by default** in every Java class.

Commonly used classes and types from the java.lang package include: String, System, Math etc.

User-defined packages : User-defined packages are those that the users define. Inside a package, you can have Java files like classes, interfaces, and a package as well (called a sub-package).

Math Class

java.lang.Math class is a built-in class. It provides mathematical functions and constants for mathematical operations.

Commonly used methods and constants:

Math.[abs](#)(a) Returns the absolute value of a value.

Math.[sqrt](#)(a) Returns the sqrt root of a double value.

Math.[ceil](#)(a) Returns the closest value that is greater than or equal to the argument

Math.[max](#)(a, b) Returns the greater of two values.

Math.[min](#)(a, b) Returns the smaller of two values.

Math.[pow](#)(a, b) Returns a raised to the power of b.

Math.[random](#)() Returns a double value with a positive sign, greater than or equal to 0.0

and less than 1.0

Math.PI, Math.floor(a)(closest value that is greater than or equal to the argument) etc.